Statistical Properties of the Sharpe Ratio Estimator

A Robustness Analysis under Various Assumptions on Returns

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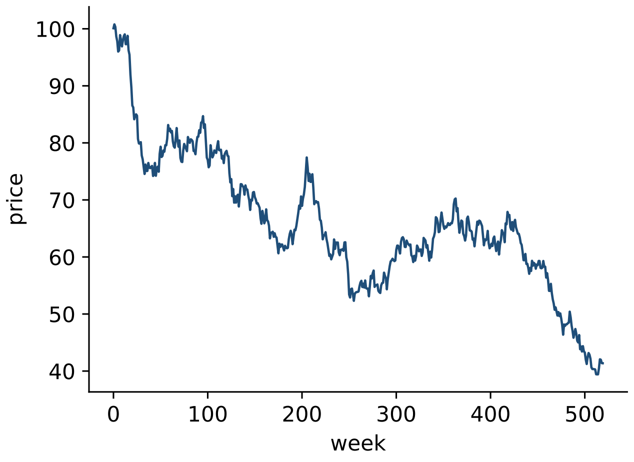
How much can you trust a Sharpe ratio?

How reliable is this number?

- A reported Sharpe ratio of $\widehat{\mathrm{SR}} = 0.5$.
- A population quantity, estimated from a finite sample:

$$\mathrm{SR} = \frac{\mu}{\sigma}, \qquad \widehat{\mathrm{SR}} = \frac{\hat{\mu}}{\hat{\sigma}}$$

- One question for the whole talk: **how much should we trust it?**

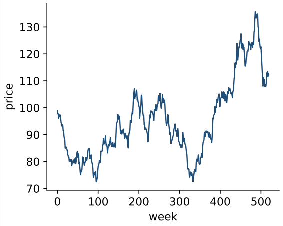


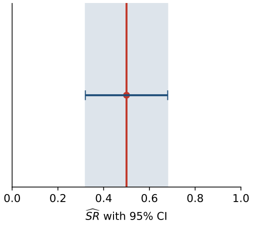
Reported $\widehat{\mathrm{SR}} = 0.5$.

Measuring reliability: the asymptotic variance

$$\sqrt{T}\,(\widehat{\mathrm{SR}} - \mathrm{SR}) \xrightarrow{d} \mathcal{N}(0,\ V_{\widehat{\mathrm{SR}}})$$

Classical IID–Normal result (Lo, 2002): $V_{\widehat{\mathrm{SR}}}^{\mathrm{IID\,N}} = 1 + \frac{1}{2}\,\mathrm{SR}^2$ CI: $\mathrm{IC}_{1-\alpha} = \left[\widehat{\mathrm{SR}} \pm z_{1-\alpha/2}\sqrt{V_{\widehat{\mathrm{SR}}}/T}\right]$

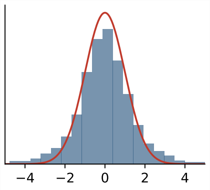




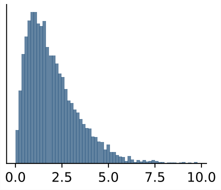
All of this rests on the IID–Normal assumption.

Real returns are not IID Normal — Cont's stylized facts

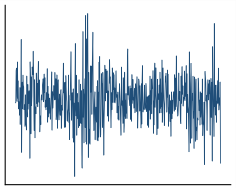
Heavy tails



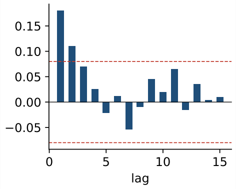
Skewness



Volatility clustering



Autocorrelation (low freq.)



Each breaks an assumption behind $1 + \frac{1}{2}\mathrm{SR}^2$.

Prior work: one variance formula per model

Model	$V_{\widehat{SR}}$
IID Normal[a]	$1 + \frac{1}{2} SR^2$
IID Non-Normal[b]	$1 - \gamma\, SR + \frac{\kappa - 1}{4} SR^2$
AR(1)[c]	$\frac{1+\phi}{1-\phi} - \frac{1+\phi+\phi^2}{1-\phi^2} \gamma SR + \frac{1+\phi^2}{1-\phi^2} \frac{\kappa-1}{4} SR^2$
GARCH(1,1)[d]	$1 - \frac{1-\beta}{1-\alpha-\beta} \gamma SR + \frac{\kappa-1}{4} \frac{(1-\beta)^2 (1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)} SR^2$

The point estimate $\widehat{\mathrm{SR}}$ is identical in every row — only the variance (interval width) changes.

γ = skewness, κ = kurtosis.

[a] (Lo 2002) [b] (Mertens 2002) [c] (López de Prado, Lipton, and Zoonekynd 2025) [d] (López de Prado, Porcu, et al. 2026)

Contribution: closing the gap — AR(1)-GARCH(1,1)

No closed form existed when both features are present: AR in the mean **and** GARCH in the variance.

Asymptotic variance, AR(1)-GARCH(1,1), general innovations

$$V_{\widehat{\mathrm{SR}}} = \frac{1+\phi}{1-\phi} - \frac{(1+\phi+\phi^2)}{(1-\phi^2)(1+2\phi\alpha-\phi\beta)}\left[2\phi\alpha + \frac{(1-\beta)(1-\phi(\alpha+\beta))}{1-\alpha-\beta}\right]\gamma\,\mathrm{SR} + \frac{\mathrm{SR}^2}{4(1-\phi^2)}\left(4\phi^2 + ((1+\phi^2)\kappa - 1 - 5\phi^2)\frac{\Sigma}{1+X} + \gamma^2 Z\left(N - \frac{M}{1+X}\Sigma\right)\right)$$

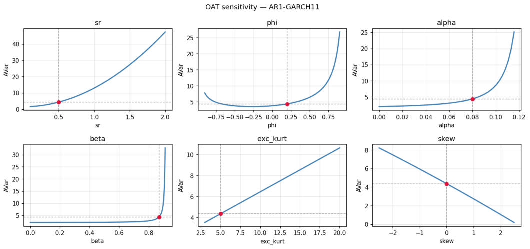
γ = skewness, κ = kurtosis.

Auxiliary terms:

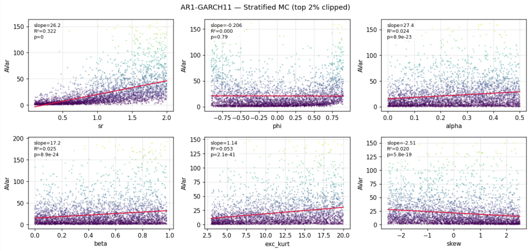
$$M = 1+\phi^2(2\alpha-\beta), \qquad N = 2\phi^2\alpha + \frac{1-\beta}{1-\alpha-\beta}\left(1-\phi^2(\alpha+\beta)\right), \qquad X = \alpha\frac{6\phi^2}{1-\phi^2(\alpha+\beta)}\frac{1-\alpha\beta-\beta^2}{1-2\alpha\beta-\beta^2}$$

$$\Sigma = \tfrac{2}{3}X + \frac{(1-\beta)^2}{(1-\alpha-\beta)^2}\frac{1-(\alpha+\beta)^2}{1-2\alpha\beta-\beta^2}, \qquad Z = \tfrac{2}{3}\alpha\frac{(1-\phi(\alpha+\beta))(1+\phi+\phi^2)^2}{(1+\phi)^2(1+2\phi\alpha-\phi\beta)^2}\frac{4\phi}{1-\phi^2(\alpha+\beta)}$$

Which parameters move the variance?



One-at-a-time sweep



Axis-stratified Monte Carlo

- SR is the dominant driver (convex, quadratic).
- Kurtosis & skewness next (steady, $\sim$linear).
- ϕ, α, β: mild on average, **explosive near the boundaries** ($\phi \to 1$, $\alpha + \beta \to 1$).

Do the confidence intervals actually work?

Testing the intervals: coverage on synthetic data

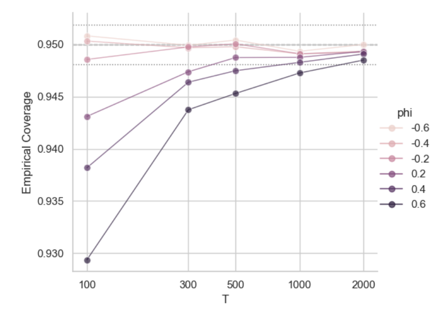
Simulate from a known DGP; measure how often the 95% interval contains the true SR.

Why coverage, not variance?

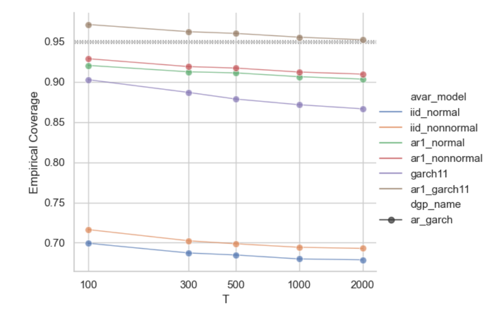
- It is the **inferential target** the user cares about.
- It is a **joint test** — catches bad variance, bias, *and* non-normality.
- It has a **calibrated target**: a binomial acceptance band around 0.95.

Curve below the band $\Rightarrow$ intervals too narrow $\Rightarrow$ overconfident.

Two findings: sign asymmetry & the AR-GARCH gap

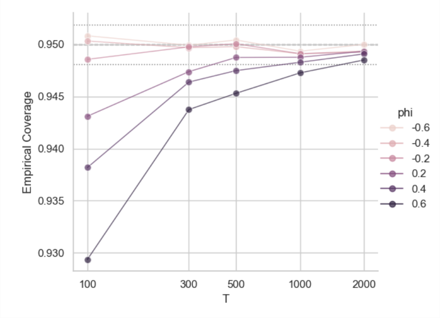


AR(1) DGP: $\rho = +0.6$ vs $\rho = -0.6$



AR-GARCH DGP: all candidate models

Two findings: sign asymmetry & the AR-GARCH gap



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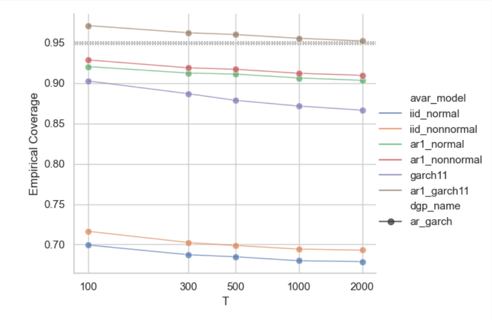
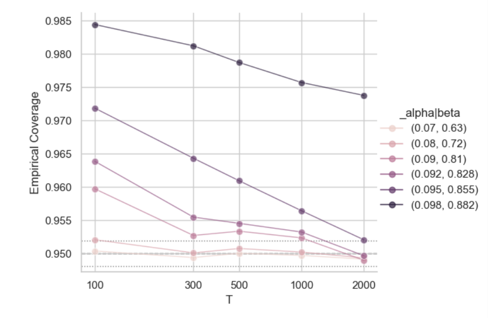


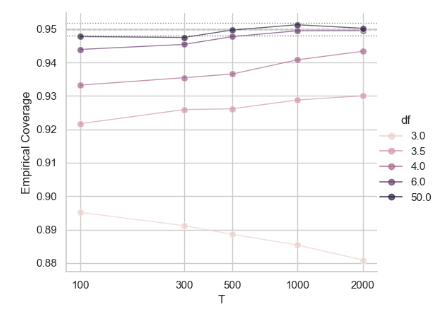
AR-GARCH DGP: all candidate models

- Autocorrelation is asymmetric in sign: positive $\rho \Rightarrow$ coverage $\approx$ 0.69; negative $\rho \Rightarrow \approx$ 0.99.
- With both stylized facts present, only the AR-GARCH formula closes the gap.

Where the asymptotics strain



GARCH persistence $\alpha + \beta$



Student-t degrees of freedom *df*

- Higher GARCH persistence $\Rightarrow$ **slower** convergence to nominal.
- No fourth moment ($df < 4$): coverage **drifts away** as T grows.
- Matches Kao et al. (2025): rate T^{φ} with $\varphi = 1 - 2/\kappa < \frac{1}{2}$ — $\sqrt{T}$ breaks down.

Is the point estimate even right?

The point estimate is biased — and a popular fix is wrong

$\widehat{\mathrm{SR}} = \hat{\mu}/\hat{\sigma}$ is biased in finite samples. Christie (2005) assumes $\hat{\sigma}$ unbiased, but by Jensen (concavity of $\sqrt{\cdot}$):

$$\mathbb{E}[\hat{\sigma}] = \mathbb{E}\left[\sqrt{\hat{\sigma}^2}\right] < \sqrt{\mathbb{E}[\hat{\sigma}^2]} \leq \sigma$$

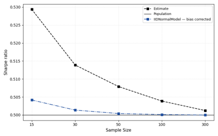
Christie: $\mathrm{SR}\left(1 + \frac{1}{2T}\right)$ — too small. Correct: $\mathrm{SR}\left(1 + \frac{3}{4T}\right)$. General correction via the same S:

$$\mathbb{E}[\widehat{\mathrm{SR}}] - \mathrm{SR} \approx \frac{\mathrm{SR}}{2(T-1)}\left(\frac{S_{11}}{\sigma^2} - 1\right) + \frac{1}{2T}\left(-\frac{S_{12}}{\sigma^3} + \frac{3\,\mathrm{SR}}{4\sigma^4} S_{22}\right)$$

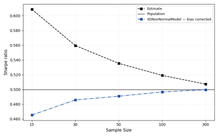
Model	Corrected estimator $\widehat{\mathrm{SR}}_{\text{corr}}$
IID Normal	$\widehat{\mathrm{SR}}\left(1 + \frac{3}{4T}\right)^{-1}$
GARCH(1,1)	$\left(\widehat{\mathrm{SR}} + \frac{\gamma}{2T}\,\frac{1-\beta}{1-\alpha-\beta}\right)\left(1 + \frac{3(\kappa-1)}{8T}\,\frac{(1-\beta)^2(1+\alpha+\beta)}{(1-\alpha-\beta)(1-2\alpha\beta-\beta^2)}\right)^{-1}$
AR(1)-GARCH(1,1)	$\left(\widehat{\mathrm{SR}} + \frac{1}{2T}\,\frac{S_{12}}{\sigma^3}\right)\left[1 + \frac{1}{2(T-1)}\left(\frac{S_{11}}{\sigma^2} - 1\right) + \frac{3}{8T\sigma^4} S_{22}\right]^{-1}$

First analytical bias corrections under GARCH and AR-GARCH.

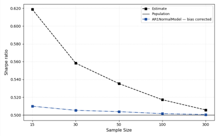
The corrections converge faster



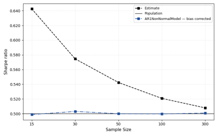
(a) IID Normal



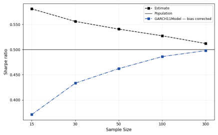
(b) IID Non-Normal



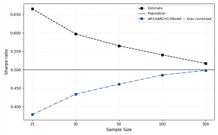
(c) AR(1) Normal



(d) AR(1) Non-Normal



(e) GARCH(1,1)



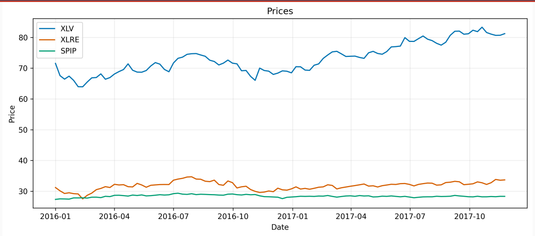
(f) AR(1)-GARCH(1,1)

Figure 1: Individual comparison of the improvement from the bias correction.

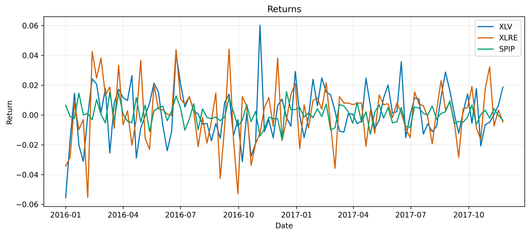
Corrected estimator (blue) converges faster in every model. GARCH is slightly off at very small T, then overtakes — reported honestly.

Does the framework survive real data?

Real data: three SPDR ETFs (2016–2025, weekly)



Price levels



Weekly log-returns

Chosen because they show both serial correlation in the level and conditional heteroskedasticity.

Series	Ljung–Box (level)	ARCH-LM	Normality (JB)	Selected specification
XLV	reject **	reject ***	reject ***	AR(1)-GARCH(1,1)-skew-t
XLRE	reject ***	reject ***	reject ***	AR(1)-GARCH(1,1)-skew-t
SPIP	reject ***	reject ***	reject ***	AR(1)-GARCH(1,1)-skew-t

Hierarchical testing (Ljung–Box $\to$ Engle LM $\to$ JB/symmetry) selects the same model for all three.

Inference across models: the Probabilistic Sharpe Ratio

$$\widehat{\mathrm{PSR}}(\mathrm{SR}^\star) = \Phi\left(\frac{\widehat{\mathrm{SR}} - \mathrm{SR}^\star}{\sqrt{V_{\mathcal{M}}(\hat{\theta})/T}}\right), \qquad \mathrm{SR}^\star = 0$$

Same $\widehat{\mathrm{SR}}$ in every column; only $V_{\mathcal{M}}$ changes.

Series	iid Normal	AR(1)	GARCH(1,1)	AR(1)-GARCH(1,1)
XLV	95.80	96.95	94.94	96.22
XLRE	76.51	79.89	76.14	78.72
SPIP	42.88	42.20	42.80	42.21

XLV crosses the 95% line; the correction is **non-monotone** in model complexity.

Comparing PSRs across columns is not a formal model test — each uses a different reference distribution. It mirrors an analyst committing to one specification.

Beyond a single number: pairing with VaR

Beyond one metric: joint distribution with Value-at-Risk

$$\sqrt{T}\begin{pmatrix}\widehat{\mathrm{SR}}-\mathrm{SR}\\ \widehat{\mathrm{VaR}}-\mathrm{VaR}\end{pmatrix}\xrightarrow{d}\mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix},\ V_h\right)$$

Non-parametric (empirical quantile)

$$V_h^{np}=\begin{pmatrix}1+\frac{\mathrm{SR}^2}{2} & \frac{\mu z_\alpha}{2}-\sigma\\ \frac{\mu z_\alpha}{2}-\sigma & \sigma^2\,\frac{\alpha(1-\alpha)}{\varphi(z_\alpha)^2}\end{pmatrix}$$

Parametric (Gaussian)

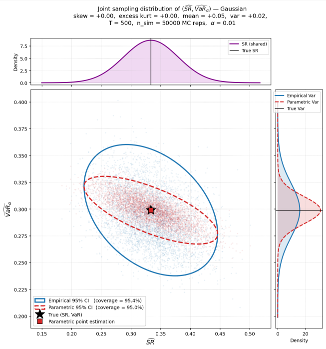
$$V_h^{p}=\begin{pmatrix}1+\frac{\mathrm{SR}^2}{2} & \frac{\mu z_\alpha}{2}-\sigma\\ \frac{\mu z_\alpha}{2}-\sigma & \sigma^2\left(1+\frac{z_\alpha^2}{2}\right)\end{pmatrix}$$

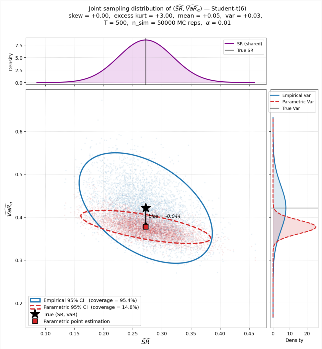
α	0.05	0.01	0.001
V_{22}^{np}/V_{22}^{p}	≈ 1.90	≈ 3.77	≈ 11

- Negative covariance ($\frac{\mu z_\alpha}{2}-\sigma<0$): a shock that raises $\widehat{\mathrm{SR}}$ also lowers $\widehat{\mathrm{VaR}}$ — they err together.
- Only the $(2,2)$ entry differs. Parametric is more efficient; the gap grows in the tail.

$\varphi(\cdot)=$ standard normal density, $z_\alpha=\Phi^{-1}(\alpha)$.

Efficiency vs robustness, in one picture





Gaussian returns: parametric ellipse **inside** the non-parametric one. Coverage 95.0% vs 95.4%.

Student-t_6 returns: parametric ellipse biased (shift -0.044), coverage 14.8%; non-parametric stays 95.4%.

Parametric wins when the distribution is right; fails badly when it is wrong.

How reliable is this number?

1. **Precision** depends on the model assumed for returns.
2. The **wrong model** makes intervals lie — usually too narrow.
3. In **small samples** the estimate is biased — the popular fix is too small.
4. Even when **correctly specified**, parameters near the **boundaries** (ϕ, $\alpha + \beta$ persistence) cause trouble.

A single framework — built on the long-run variance matrix S — answers all four for the AR-GARCH world real returns live in.

Thank you — questions welcome.

Code: `github.com/Fabro2701/sharpe_ratio_TFM`